

Class09 Halloween Candy

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Background

Today we are taking a detour to analyzze a fun data-set with the most useful analysis method we have learned so far- Principal Component Analysis (PCA)

Data Import

The data is all related to Halloween candy and is from the 538 website

```
candy <- read.csv("candy-data.txt", row.names = 1)
head(candy)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer
100 Grand	1	0	1	0	0	1
3 Musketeers	1	0	0	0	1	0
One dime	0	0	0	0	0	0
One quarter	0	0	0	0	0	0

Air Heads	0	1	0	0	0	0
Almond Joy	1	0	0	1	0	0
	hard	bar	pluribus	sugarpercent	pricepercent	winpercent
100 Grand	0	1	0	0.732	0.860	66.97173
3 Musketeers	0	1	0	0.604	0.511	67.60294
One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

Q1. How many different candy types are in this dataset?

```
nrow(candy)
```

```
[1] 85
```

```
dim(candy)
```

```
[1] 85 12
```

There are 85 different candy types in this dataset.

Q2. How many fruity candy types are in the dataset?

```
sum(candy$fruity == 1)
```

```
[1] 38
```

There are 38 fruity candy types in the dataset.

What's my favorite candy

```
library(dplyr)
```

```
Attaching package: 'dplyr'
```

The following objects are masked from 'package:stats':

```
filter, lag
```

The following objects are masked from 'package:base':

```
intersect, setdiff, setequal, union
```

```
candy |>
  filter(row.names(candy)=="Nerds") |>
  select(winpercent)
```

```
      winpercent
Nerds    55.35405
```

Q3. What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value?

My favorite candy is Nerds and it's winpercent value is 55.35405.

Q4. What is the winpercent value for "Kit Kat"?

```
library(dplyr)

candy |>
  filter(row.names(candy)=="Kit Kat") |>
  select(winpercent)
```

```
      winpercent
Kit Kat    76.7686
```

The winpercent value for Kit Kat is 76.7686

What is the winpercent value for "Tootsie Roll Snack Bars"?

```
library(dplyr)

candy |>
  filter(row.names(candy)=="Tootsie Roll Snack Bars") |>
  select(winpercent)
```

```

                                winpercent
Tootsie Roll Snack Bars      49.6535

```

The winpercent of Tootsie Roll Snack Bars is 49.6535.

Side-note: There is a useful `skim()` function in the `skimr` package that can help give you a quick overview of a given dataset. Let's install this package and try it on our candy data.

```

library("skimr")
skim(candy)

```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyal- mondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedrice- wafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

The winpercent variable looks to be on a different scale compared to the other variables in the dataset. Most of the other columns have values between zero and one for their means while winpercent has much higher values.

Q7. What do you think a zero and one represent for the candy\$chocolate column?

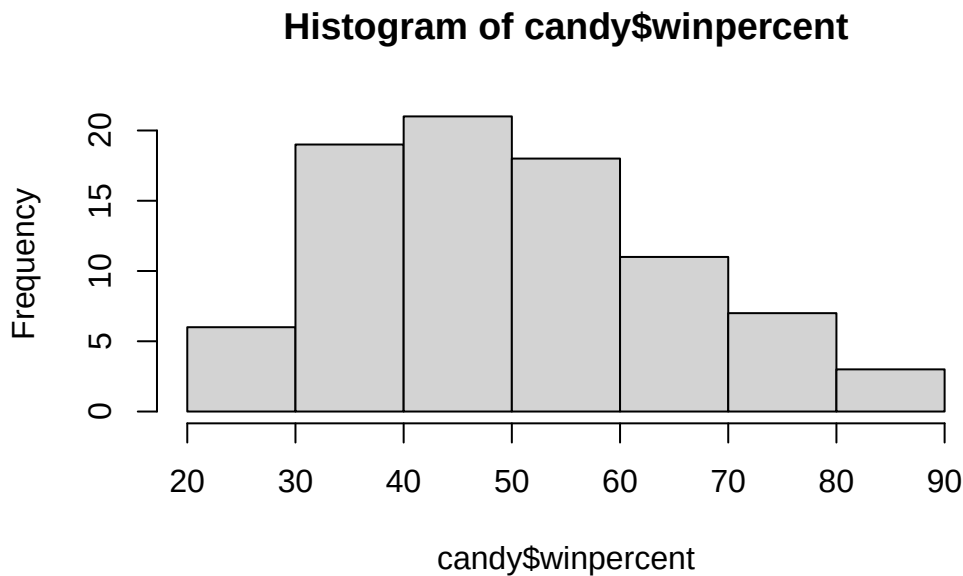
I think that a value of 0 likely means that the candy does not contain chocolate while a value of 1 means the candy does have chocolate.

Exploratory Analysis

A good place to start any exploratory analysis is with a histogram. You can do this most easily with the base R function `hist()`. Alternatively, you can use `ggplot()` with `geom_hist()`. Either works well in this case and (as always) it's your choice.

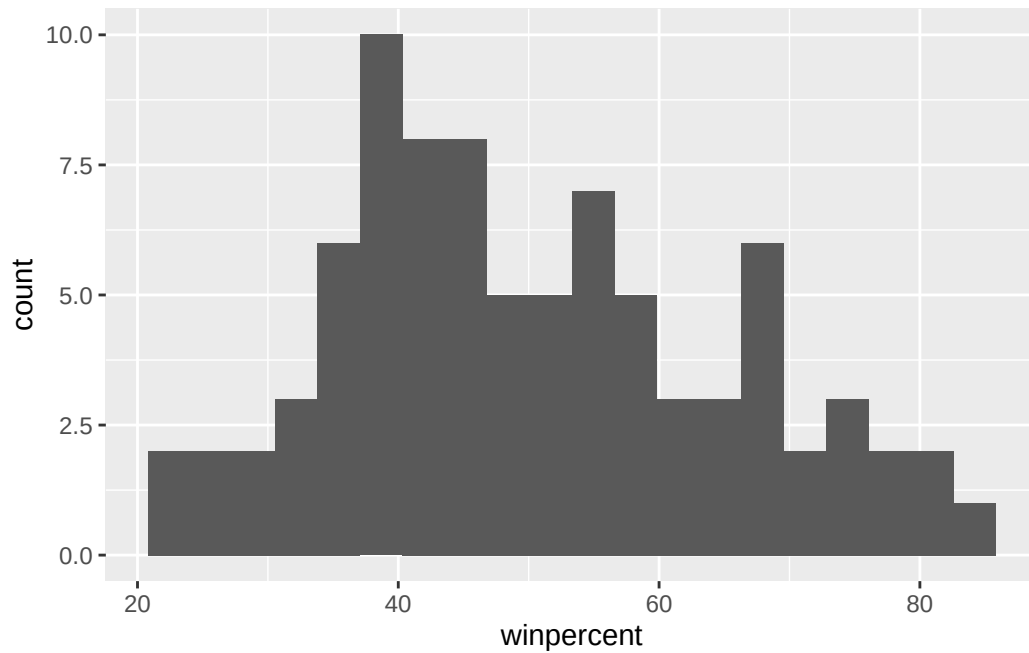
Q8. Plot a histogram of winpercent values using both base R and ggplot2.

```
hist(candy$winpercent)
```



```
library(ggplot2)

ggplot(candy)+
  aes(winpercent)+
  geom_histogram(bins=20)
```



Q9. Is the distribution of winpercent values symmetrical?

No, the distribution is not symmetrical.

Q10. Is the center of the distribution above or below 50%?

```
summary(candy$winpercent)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
22.45	39.14	47.83	50.32	59.86	84.18

The center of the distribution is likely below 50%. This is indicated by the median at 47.83 and the fact that the histograms above are skewed.

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
aggregate(winpercent ~ chocolate, data = candy, mean)
```

```
  chocolate winpercent
1          0  42.14226
2          1  60.92153
```

```
aggregate(winpercent ~ fruity, data = candy, mean)
```

```
  fruity winpercent
1       0  55.32712
2       1  44.11974
```

On average chocolate candy is ranked higher than fruit candy. Chocolate candies have an average winpercent of 60.92% while fruity candies have an average winpercent of 44.12%.

Q12. Is this difference statistically significant?

```
# Compare winpercent of chocolate vs fruity candy

chocolate <- candy$winpercent[as.logical(candy$chocolate)]
fruity <- candy$winpercent[as.logical(candy$fruity)]

t.test(chocolate, fruity)
```

Welch Two Sample t-test

```
data: chocolate and fruity
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

Yes, this difference is statistically significant. The two sample t-test produced a p-value of 2.871e-08 which is far below 0.05. This indicates that the average winpercent for chocolate candy is significantly higher than that of fruit candy.

Overall Candy Rankings

Q13. What are the five least liked candy types in this set?

```
head(candy[ order(candy$winpercent), ], n=5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Nik L Nip	0	1	0		0	0
Boston Baked Beans	0	0	0		1	0
Chiclets	0	1	0		0	0
Super Bubble	0	1	0		0	0
Jawbusters	0	1	0		0	0

	crisped	rice	wafer	hard	bar	pluribus	sugar	percent	price	percent
Nik L Nip				0	0	0	1	0.197		0.976
Boston Baked Beans				0	0	0	1	0.313		0.511
Chiclets				0	0	0	1	0.046		0.325
Super Bubble				0	0	0	0	0.162		0.116
Jawbusters				0	1	0	1	0.093		0.511

	winpercent
Nik L Nip	22.44534
Boston Baked Beans	23.41782
Chiclets	24.52499
Super Bubble	27.30386
Jawbusters	28.12744

The 5 least liked candy types are: Nik L Nip, Boston Baked Beans, Chiclets, Super Bubble, and Jawbusters.

Q14. What are the top 5 all time favorite candy types out of this set?

```
head(candy[order(candy$winpercent, decreasing = TRUE), ], 5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Reese's Peanut Butter cup	1	0	0		1	0
Reese's Miniatures	1	0	0		1	0
Twix	1	0	1		0	0
Kit Kat	1	0	0		0	0
Snickers	1	0	1		1	1

	crisped	rice	wafer	hard	bar	pluribus	sugar	percent
Reese's Peanut Butter cup				0	0	0	0	0.720
Reese's Miniatures				0	0	0	0	0.034

Twix	1	0	1	0	0.546
Kit Kat	1	0	1	0	0.313
Snickers	0	0	1	0	0.546

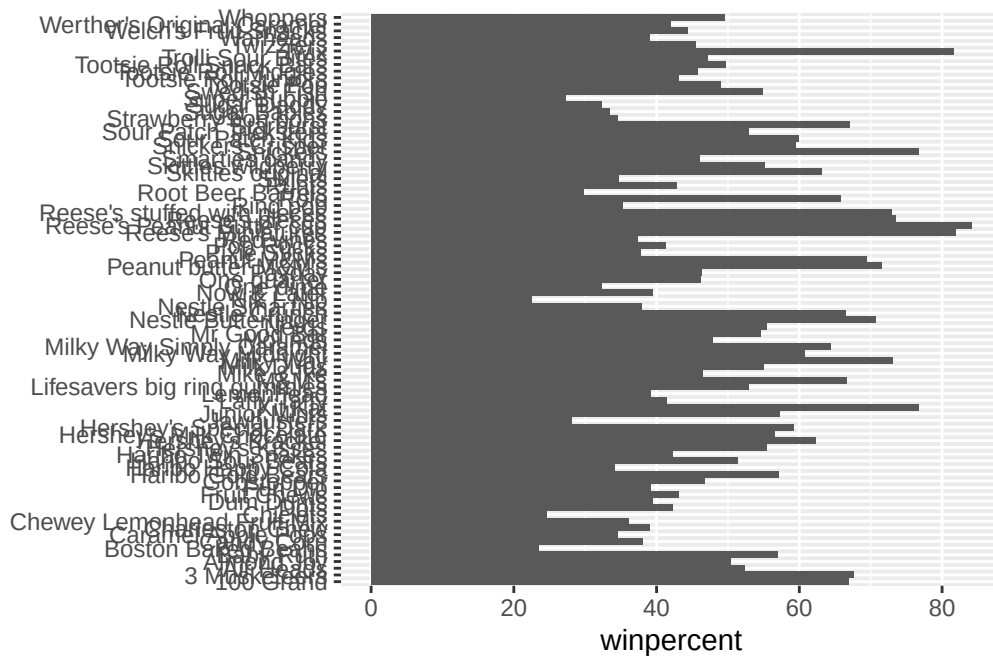
	pricepercent	winpercent
Reese's Peanut Butter cup	0.651	84.18029
Reese's Miniatures	0.279	81.86626
Twix	0.906	81.64291
Kit Kat	0.511	76.76860
Snickers	0.651	76.67378

The top 5 all time favorites are :Reese's Peanut Butter cup, Reese's Miniatures, Twix, Kit Kat, and Snickers

Q15. Make a first barplot of candy ranking based on winpercent values.

```
library(ggplot2)

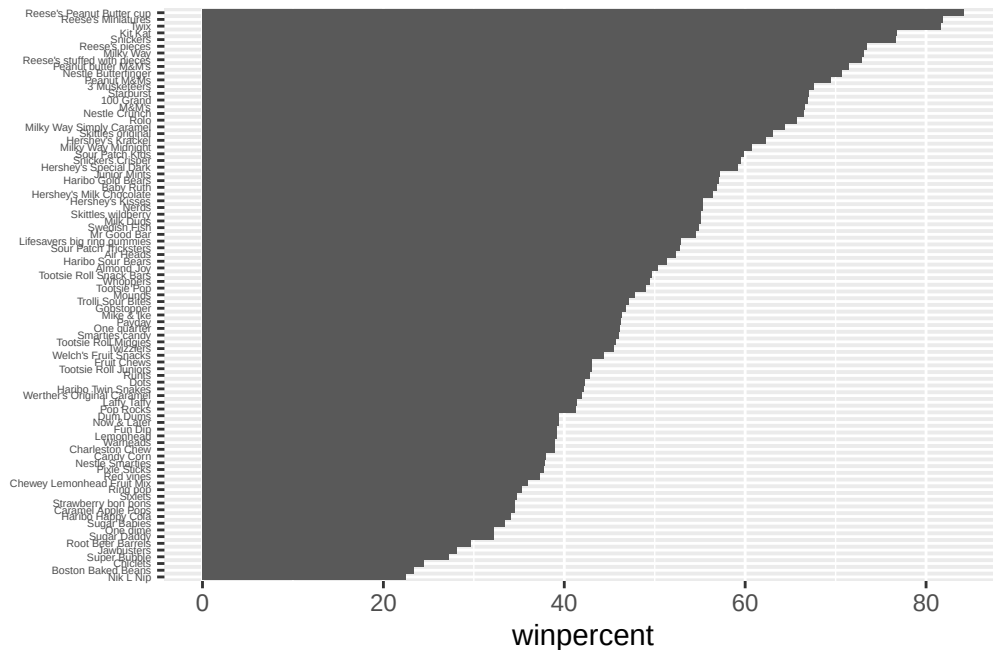
ggplot(candy) +
  aes(winpercent, rownames(candy)) +
  geom_col()+
  ylab("")
```



Q16. This is quite ugly, use the `reorder()` function to get the bars sorted by `winpercent`?

```
library(ggplot2)

ggplot(candy, aes(winpercent, reorder(rownames(candy), winpercent))) +
  geom_col() +
  ylab("") +
  theme(axis.text.y = element_text(size = 4))
```



Now let's setup a color vector (that signifies candy type) that we can then use for some future plots. We start by making a vector of all black values (one for each candy). Then we overwrite chocolate (for chocolate candy), brown (for candy bars) and red (for fruity candy) values.

```
mycols <- rep("black", nrow(candy))

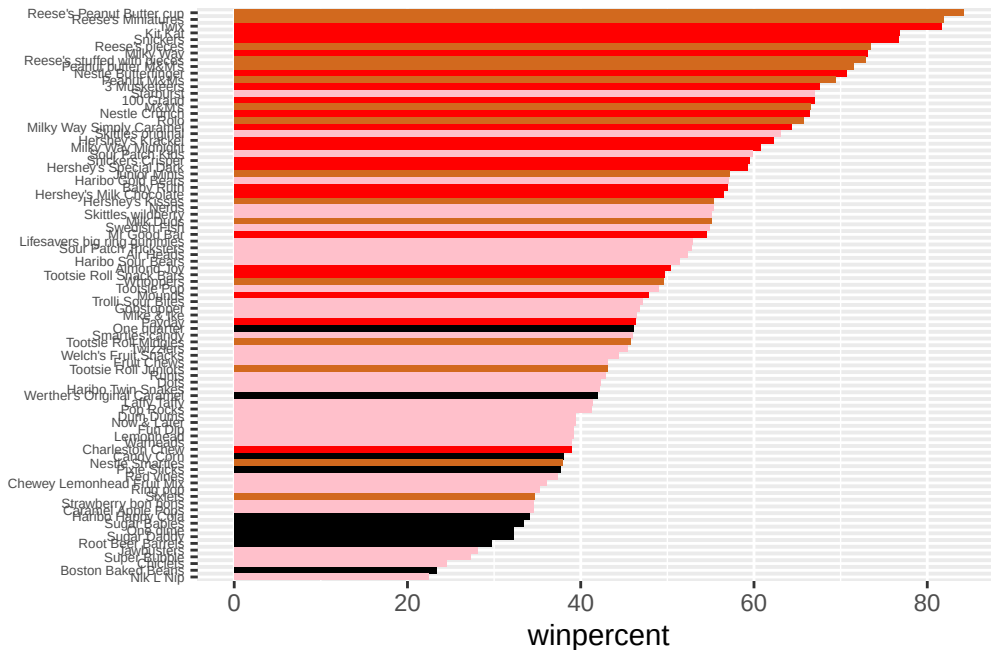
#Chocolate candy in chocolate color
mycols[ as.logical(candy$chocolate)] <- "chocolate"

# candy bars in red
mycols[ as.logical(candy$bar)] <- "red"

# fruity candy in pink
```

```
mycols[ as.logical(candy$fruity)] <- "pink"
```

```
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) +
  geom_col(fill=mycols) +
  ylab("") +
  theme(axis.text.y = element_text(size = 5))
```



Q17. What is the worst ranked chocolate candy?

According to the new plot, the worst ranked chocolate candy is Sixlets.

Q18. What is the best ranked fruity candy?

According to the new plot, the best ranked fruity candy is Starburst.

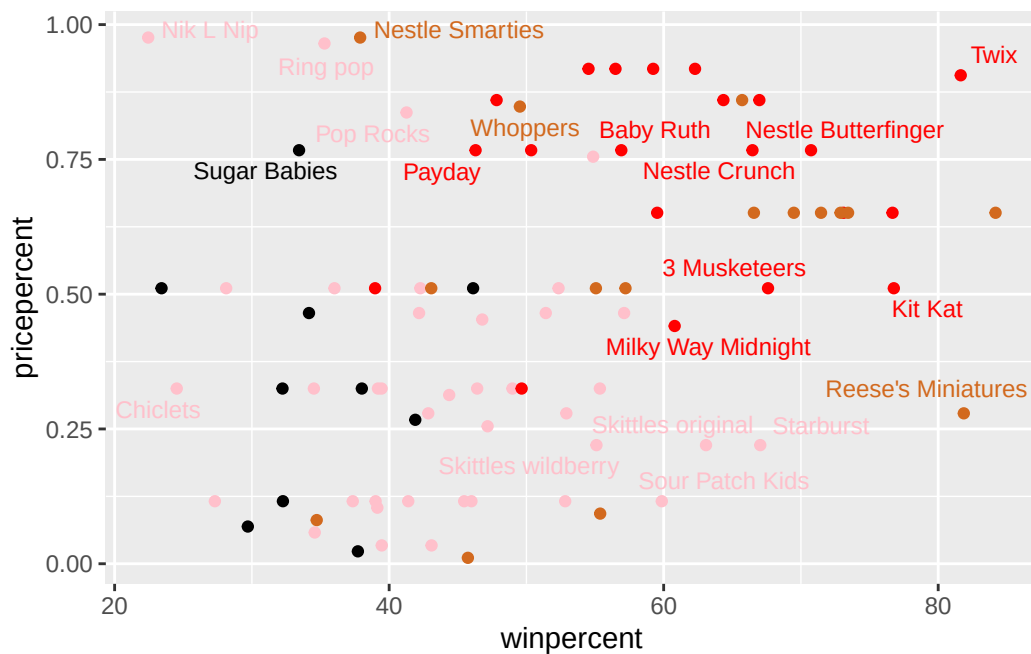
Taking a look at pricepercent

What about value for money? What is the best candy for the least money? One way to get at this would be to make a plot of `winpercent` vs the `pricepercent` variable. The `pricepercent` variable records the percentile rank of the candy's price against all the other candies in the dataset. Lower values are less expensive and higher values are more expensive.

To this plot we will add text labels so we can more easily identify a given candy. There is a regular `geom_label()` that comes with `ggplot2`. However, as there are quite a few candies in our dataset lots of these labels will be overlapping and hard to read. To help with this we can use the `geom_text_repel()` function from the **ggrepel** package.

```
library(ggrepel)

# How about a plot of win vs price
ggplot(candy) +
  aes(y= pricepercent, x=winpercent, label=rownames(candy)) +
  geom_point(col=mycols) +
  geom_text_repel(col=mycols, size=3.3, max.overlaps = 5)
```



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

```
ord <- order(candy$winpercent / candy$pricepercent, decreasing = TRUE)
head(candy[ord, c(11, 12)], n = 5)
```

	pricepercent	winpercent
Tootsie Roll Midgies	0.011	45.73675
Pixie Sticks	0.023	37.72234

Fruit Chews	0.034	43.08892
Dum Dums	0.034	39.46056
Strawberry bon bons	0.058	34.57899

Tootsie Roll Midgies is the highest ranked in terms of winpercent for the least money.

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

```
#Most Expensive
ord <- order(candy$pricepercent, decreasing = TRUE)
head(candy[ord, c(11, 12)], n = 5)
```

	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Nestle Smarties	0.976	37.88719
Ring pop	0.965	35.29076
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

```
# Least popular among most expensive
top5 <- head(candy[ord, ], n = 5)
top5[which.min(top5$winpercent), c(11, 12)]
```

	pricepercent	winpercent
Nik L Nip	0.976	22.44534

The top 5 most expensive candy types are Nik L Nip, Nestle Smarties, Ring pop, Hershey's Krackel, and Hershey's Milk Chocolate. The least popular is Nik L Nip.

Exploring the correlation structure

Before we dive into PCA in the next section, it's worth pausing to look more closely at the correlations between all pairs of variables.

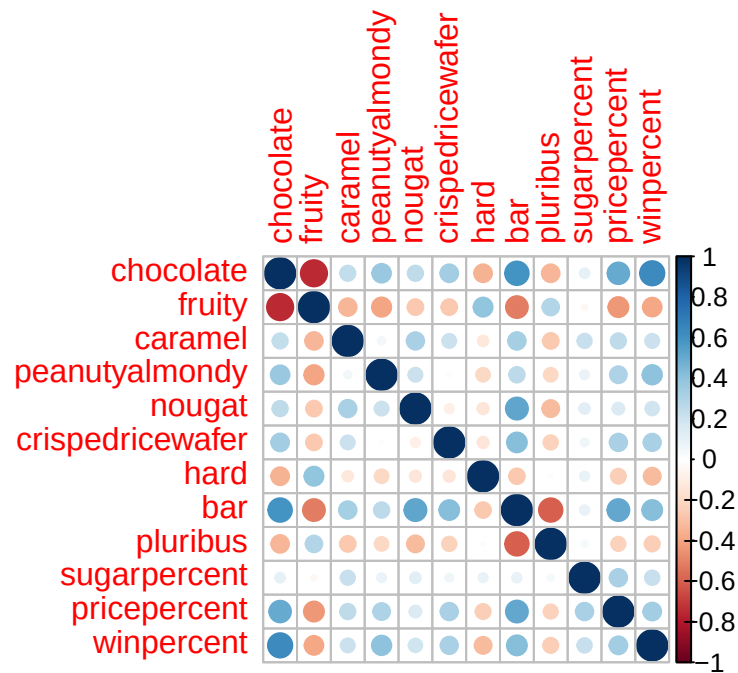
We have already seen that being a chocolate candy tends to imply being relatively expensive and popular - in other words, these variables are positively correlated with one another. Here we will examine the magnitude and direction of all variable pair combinations to see how they relate to each other more broadly.

The code below calculates the pairwise correlation between all variables and views the results with the **corrplot** package to plot a correlation matrix.

```
library(corrplot)
```

```
corrplot 0.95 loaded
```

```
cij <- cor(candy)  
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

The two variables that appear the most anti-correlated are chocolate and fruity meaning candies that contain chocolate are usually not fruity and vice versa. Another set of variables that are anti-correlated is pluribus and bar.

Q23. Use your corrplot result to make a prediction: which variables do you expect will have the largest contributions (i.e. loadings) to PC1 (i.e., drive the most separation between candies along the first principal component)?

Based on the corrplot I believe that the variables that will have the largest contributions to PC1 are chocolate, fruity, bar, and pluribus because they show strong negative correlations with other variables.

Principal Component Analysis

Let's apply PCA using the `prcomp()` function to our candy dataset remembering to set the `scale=TRUE` argument.

```
pca <- prcomp(candy, scale= TRUE)
summary(pca)
```

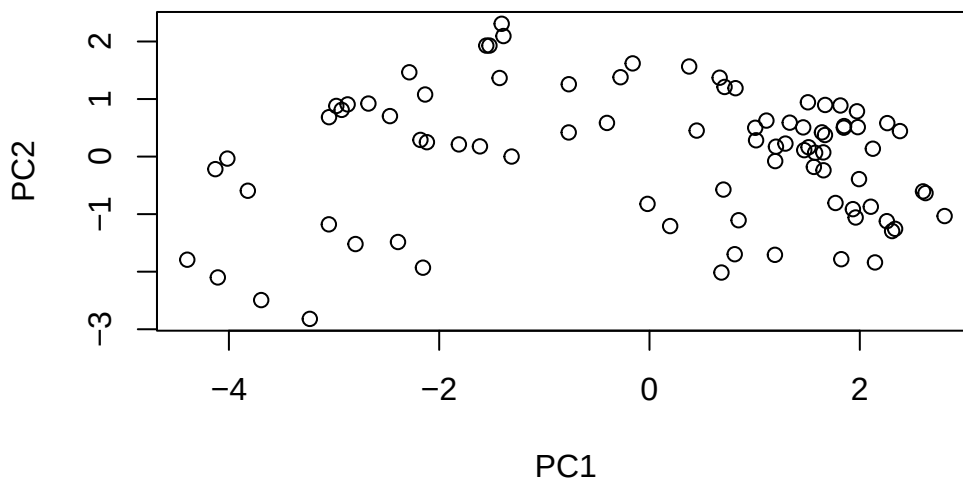
Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369

	PC8	PC9	PC10	PC11	PC12
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000

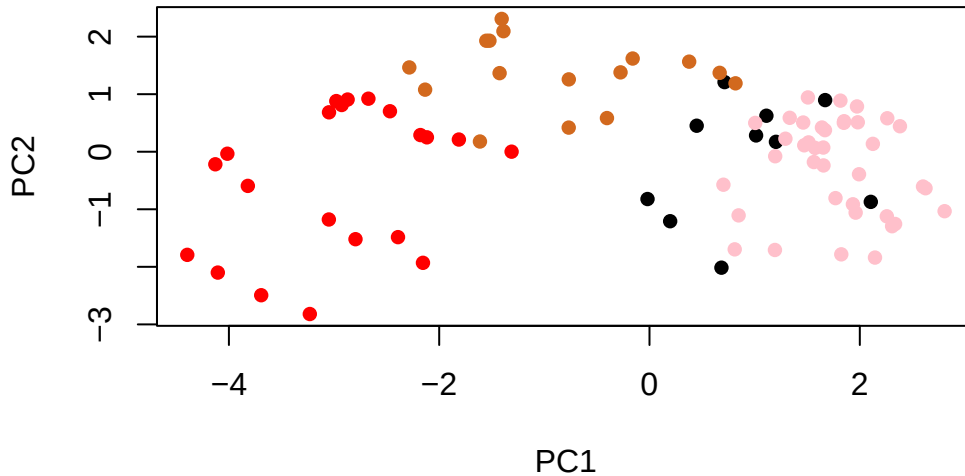
Now we can plot our main PCA score plot of PC1 vs PC2:

```
plot(pca$x[,c(1:2)])
```



We can change the plotting character and add some color:

```
plot(pca$x[,1:2], col=mycols, pch=16)
```

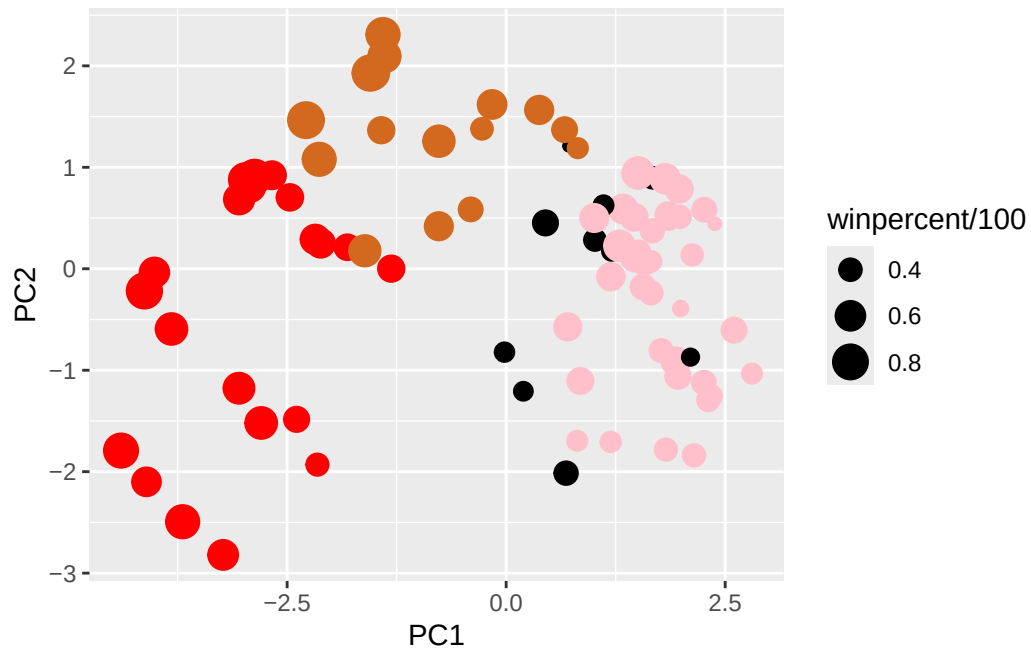


We can make a much nicer plot with the **ggplot2** package but it is important to note that ggplot works best when you supply an input data.frame that includes a separate column for each of the aesthetics you would like displayed in your final plot. To accomplish this we make a new data.frame here that contains our PCA results with all the rest of our candy data. We will then use this for making plots below

```
# Make a new data-frame with our PCA results and candy data
my_data <- cbind(candy, pca$x[,1:3])
```

```
p <- ggplot(my_data) +
  aes(x = PC1, y = PC2,
      size = winpercent/100,
      text = rownames(my_data),
      label = rownames(my_data)) +
  geom_point(col = mycols)
```

p



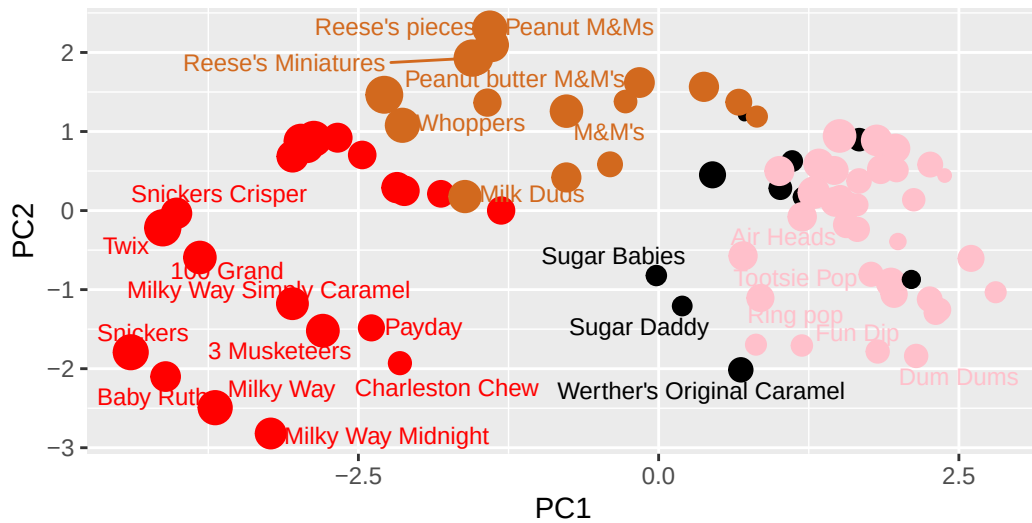
Use the **ggrepel** package to label the plot with non overlapping candy names. We will also add a title and subtitle:

```
library(ggrepel)

p + geom_text_repel(size=3.3, col=mycols, max.overlaps = 7) +
  theme(legend.position = "none") +
  labs(title="Halloween Candy PCA Space",
       subtitle="Colored by type: chocolate bar (dark brown), chocolate other (light brown),
       caption="Data from 538")
```

Halloween Candy PCA Space

Colored by type: chocolate bar (dark brown), chocolate other (light brown),



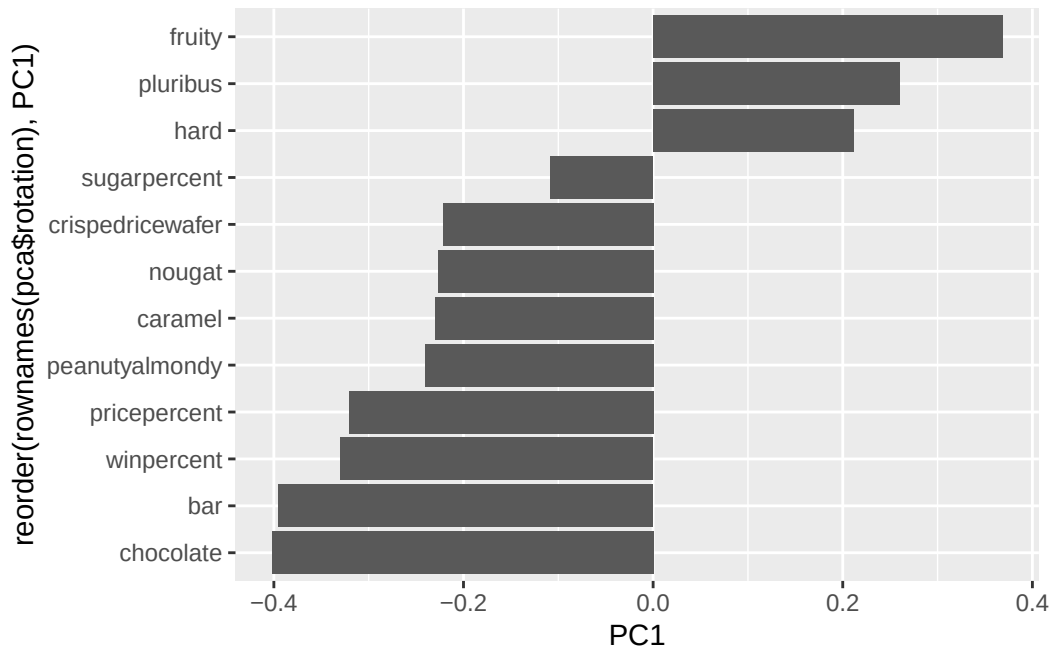
Data from 538

If you want to see more candy labels you can change the `max.overlaps` value to allow more overlapping labels or pass the ggplot object `p` to **plotly** like so to generate an interactive plot that you can mouse over to see labels:

```
##library(plotly)
##ggplotly(p)
```

Let's finish by taking a quick look at PCA our loadings. Do these make sense to you? Notice the opposite effects of `chocolate` and `fruity` and the similar effects of `chocolate` and `bar` (i.e. we already know they are correlated).

```
ggplot(pca$rotation) +
  aes(x = PC1, y = reorder(rownames(pca$rotation), PC1)) +
  geom_col()
```



Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

The variables picked up strongly by PC1 in the positive direction are fruity, pluribus, and hard. This makes sense because these features tend to go together in many non-chocolate candies. We also saw this relationship indicated earlier in the corplot where fruity was negatively correlated with chocolate and bar candies.

Summary

Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a “winning” candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

To make a winning candy it should probably be chocolate and not fruity in a bar form. These traits are associated with higher price values. The exploratory plots showed that many of the candies with high **winpercent** were chocolate bars. The correlation analysis showed that chocolate, bar, and pricepercent to be positively associated with popularity. The PCA results also further support this since PC1 separated chocolate, bar candies from fruity, pluribus and hard candies. These different analyses support each other in this conclusion.